

YOUR OFE 2025 REPORT

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2025 Cropping Season

Field Ntrials | N rates: 0, 30, 90, 150 lb N/ac | **Research question:** Can applying cover crops reduce the amount of nitrogen required to maximize grain yields?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen rate trial conducted in the Ntrials field. Four nitrogen rates were tested: 0 (Zero), 30 (Low), 90 (Standard rate), and 150 (High) lb N/ac, in a randomized complete block design with 4 blocks. Statistical comparisons use a randomized complete block design (RCBD) ANOVA with Tukey post-hoc tests. Letters in the figures indicate treatment groups that are significantly different from each other ($p < 0.05$).



Figure 1. Treatment layout.

Take-home message: Yield responded to nitrogen up to the standard rate, and not beyond it. Strips with no nitrogen yielded about 24 bu/ac less than the standard 90 lb N/ac rate, while going from 90 to 150 lb N/ac added nothing. Plant color, tissue nitrogen and end-of-season stalk nitrate did not differ among rates. Based on this data, the standard rate was enough this season and the high rate did not pay off. Continuing with cover crops remains a smart long-term investment to keep building soil health.

Results Summary

Metric	ANOVA p-value	Zero (0 lb N)	Low (30 lb N)	Standard (90 lb N)	High (150 lb N)
 Corn Yield (bu/ac)	p = 0.0223	146.2	163.1	170	170.5
 NBI (Nitrogen Balance Index)	p = 0.898	86.7	87	91.6	85.7
 Chlorophyll Index	p = 0.859	46	48.2	48.1	48
 Biomass %N (C:N analysis)	p = 0.764	4.1	4.2	4.1	4.2
 CSNT (Cornstalk Nitrate)	p = 1	4588	4778	4818	4629

Means shown. Letters from Tukey HSD test shown in individual section figures. ANOVA: RCBD, 4 blocks × 4 treatments.

Corn Yield

We measured corn yield in each strip and matched it to the nitrogen (N) rate used. Then we **compared the average yields** across the four N rates (0, 30, 90, and 150 lb N) to see if any rate clearly outperformed the others. We also **grouped the results** to show which N rates are **truly different from each other**, versus differences that are **just normal field variation** (block design-RCBD). Strips sharing the same letter are statistically the same (Tukey HSD). Finally, we tested whether yield increased steadily with more N applied (linear regression).

ZERO VS HIGH (150 LB N/AC)

146.2 ↑ +16.6% to 170.5 bu/ac

from 0 to 150 lb N/ac applied

ANOVA P-VALUE

p = 0.0223

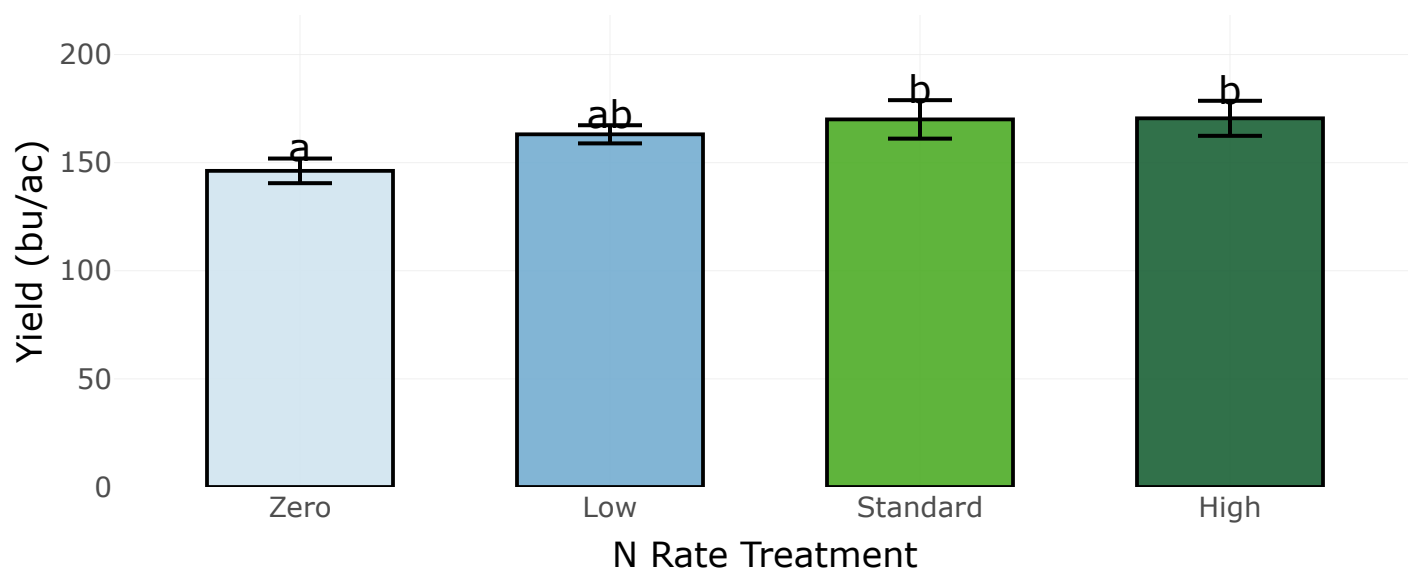
Linear regression $R^2 = 0.282$ · p = 0.0344

✓ Significant difference among treatments

Mean ± SE + Tukey

[Raw data by strip](#)

Corn Yield - Mean $\pm$ SE



Nitrogen Balance Index (NBI)

The **Nitrogen Balance Index (NBI)** is measured with the Dualex sensor and gives an idea of how much nitrogen the plant has. It compares leaf greenness to stress signals in the leaf, so **higher numbers mean the plant has enough nitrogen**. We took two leaf readings per plant and averaged them. Then we compared the average NBI across the four N rates (0, 30, 90, and 150 lb N) **to see if adding more nitrogen improved plant status**. We also grouped the results to show which rates are truly different, versus differences that are just normal field variation (block design-RCBD). Strips sharing the same letter are statistically the same (Tukey HSD).

ZERO VS HIGH (150 LB N/AC)

86.7 $\downarrow$ -1.2% to 85.7

from 0 to 150 lb N/ac applied

ANOVA P-VALUE

p = 0.898

RCBD ANOVA across Zero, Low, Standard, High

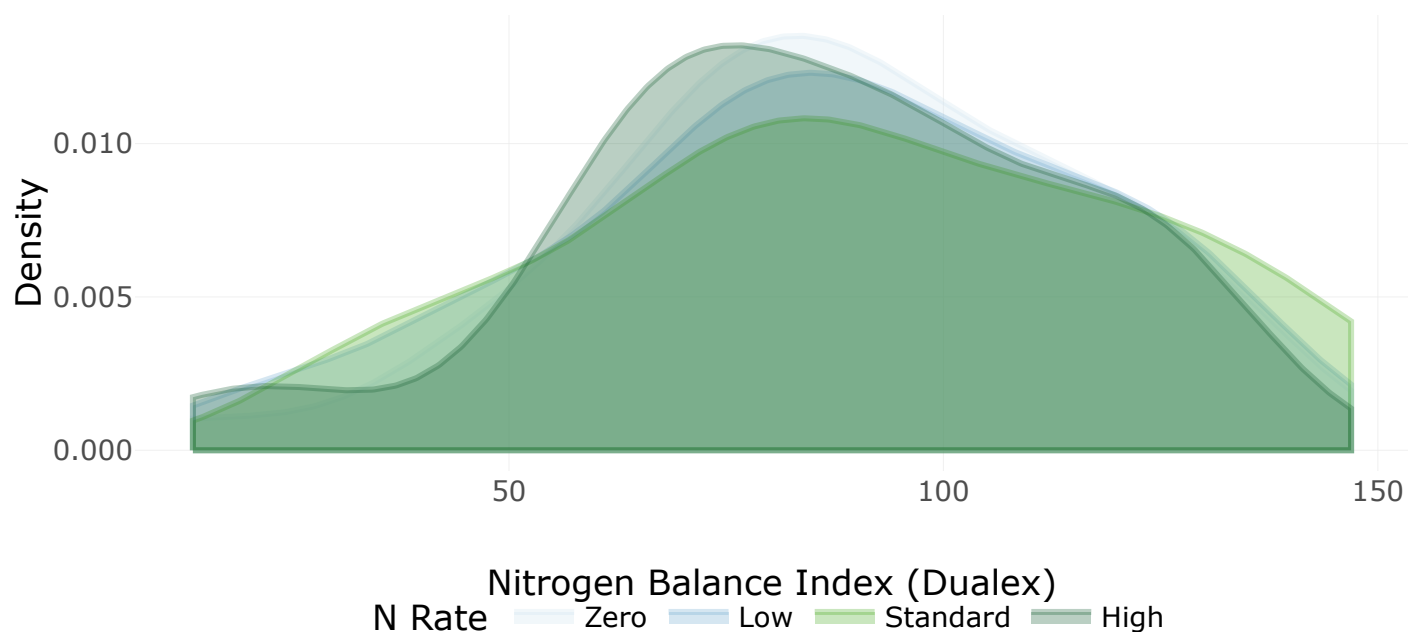
— No significant difference detected

Density

Mean $\pm$ SE + Tukey

Boxplot (raw data)

NBI Distribution by N Rate



Chlorophyll (Chl)

The **Chlorophyll Index (Chl)** estimates chlorophyll content in the leaf, a reliable proxy for nitrogen sufficiency. Greener leaves generally mean more nitrogen in the plant. As with NBI, two leaves per plant were averaged. We also grouped the results to show which rates are truly different, versus differences that are just normal field variation (block design-RCBD). Strips sharing the same letter are statistically the same (Tukey HSD).

ZERO VS HIGH (150 LB N/AC)

46 ↑ +4.4% to 48

from 0 to 150 lb N/ac applied

ANOVA P-VALUE

p = 0.859

RCBD ANOVA across Zero, Low, Standard, High

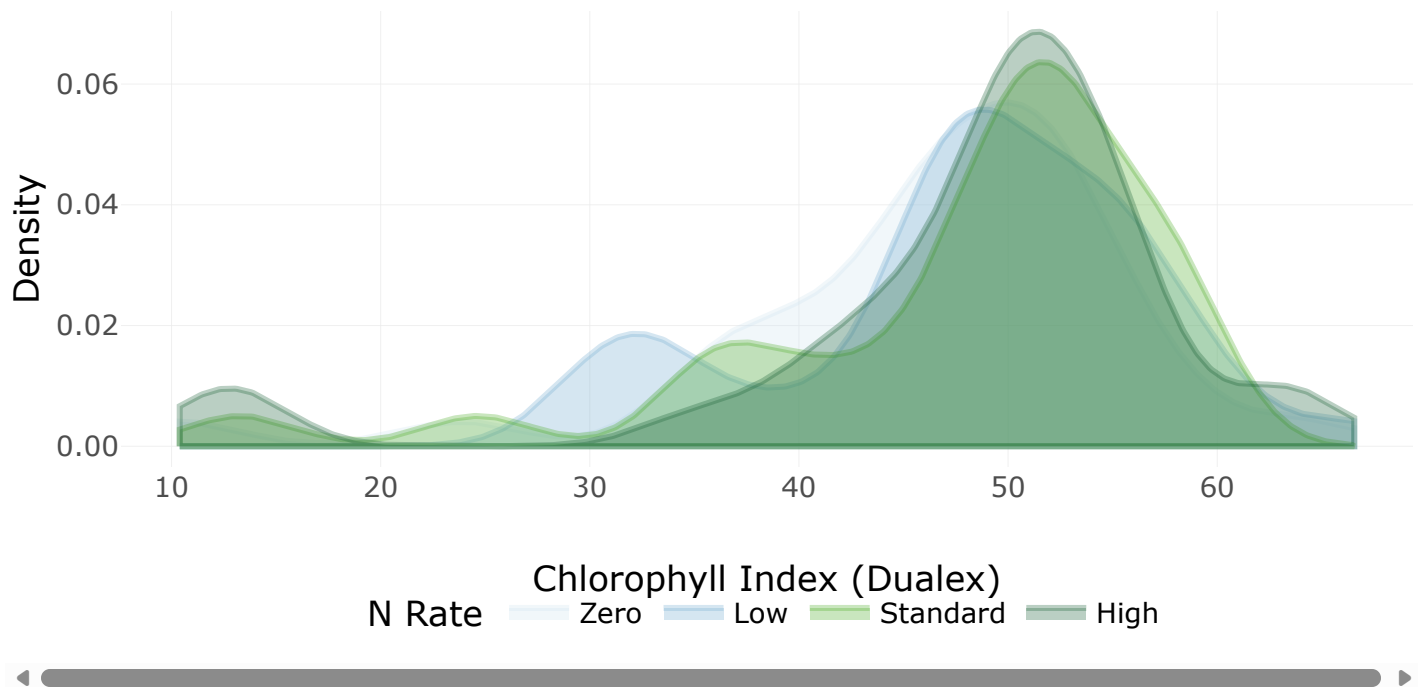
— No significant difference detected

Density

Mean ± SE + Tukey

Boxplot (raw data)

Chlorophyll Distribution by N Rate



C:N in Corn Biomass

We collected corn plant samples from each plot to measure how much nitrogen actually ended up inside the plant tissue. This is one of the most direct measurements we have, it tells you what the crop actually absorbed, not just what was available in the soil. Biomass data were available for 15 of the 16 plots, with up to two sub-samples per plot averaged together, and we used the same block design as the other analyses to keep the comparison fair. We also ran a linear regression to see if tissue nitrogen climbed steadily with more N applied.

ZERO VS HIGH (150 LB N/AC)

4.1 ↑ +2.1% to 4.2 %N

from 0 to 150 lb N/ac applied

ANOVA P-VALUE

p = 0.764

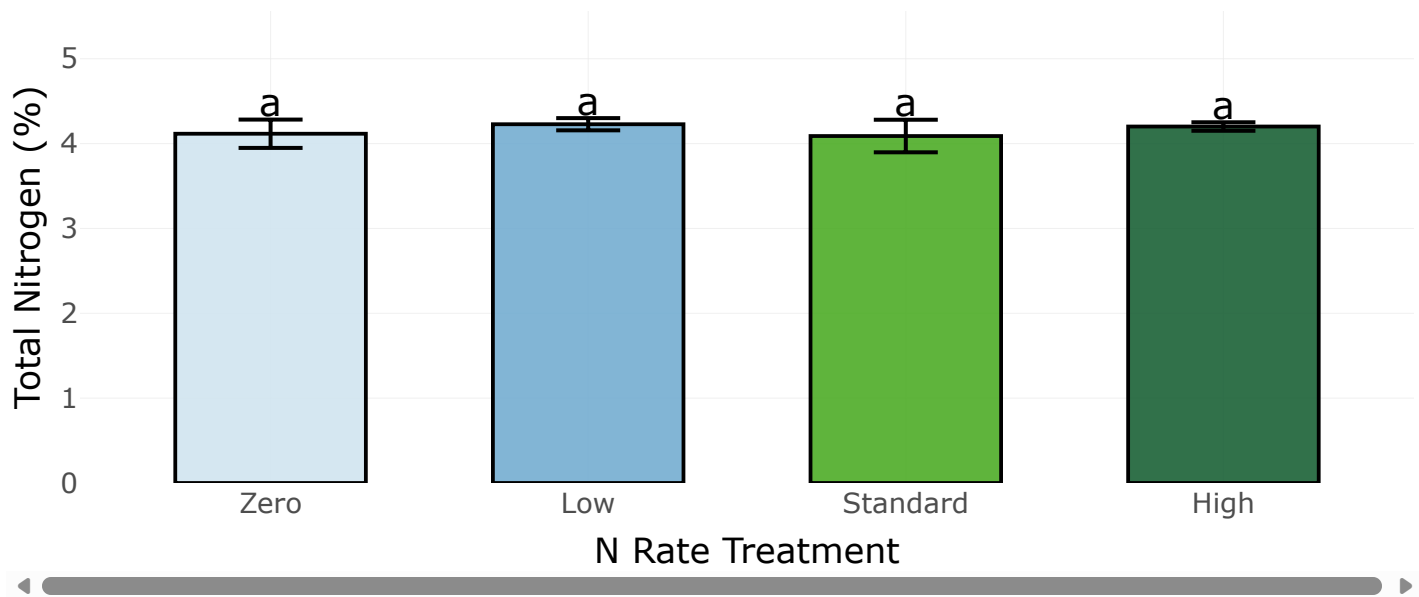
Linear regression $R^2 = 0.003$ · p = 0.782

— No significant difference detected

Mean ± SE + Tukey

[Raw data points](#)

Nitrogen in Corn Biomass (Mean %N $\pm$ SE)



Cornstalk Nitrate Test (CSNT)

The Cornstalk Nitrate Test (CSNT) measures nitrate concentration in the lower cornstalk at the end of the season. It is a post-harvest diagnostic tool: values **below 250 ppm indicate insufficient nitrogen**, while the **optimal range is between 750 and 2000 ppm**. Each plot had two samples (E and W positions), giving **32 observations** across the 16 plots. The red dashed line marks the 250 ppm deficiency threshold and the green line the 750 ppm optimal threshold.

ZERO VS HIGH (150 LB N/AC)

4588 $\uparrow$ +0.9% to 4629 ppm

from 0 to 150 lb N/ac applied

ANOVA P-VALUE

p = 1

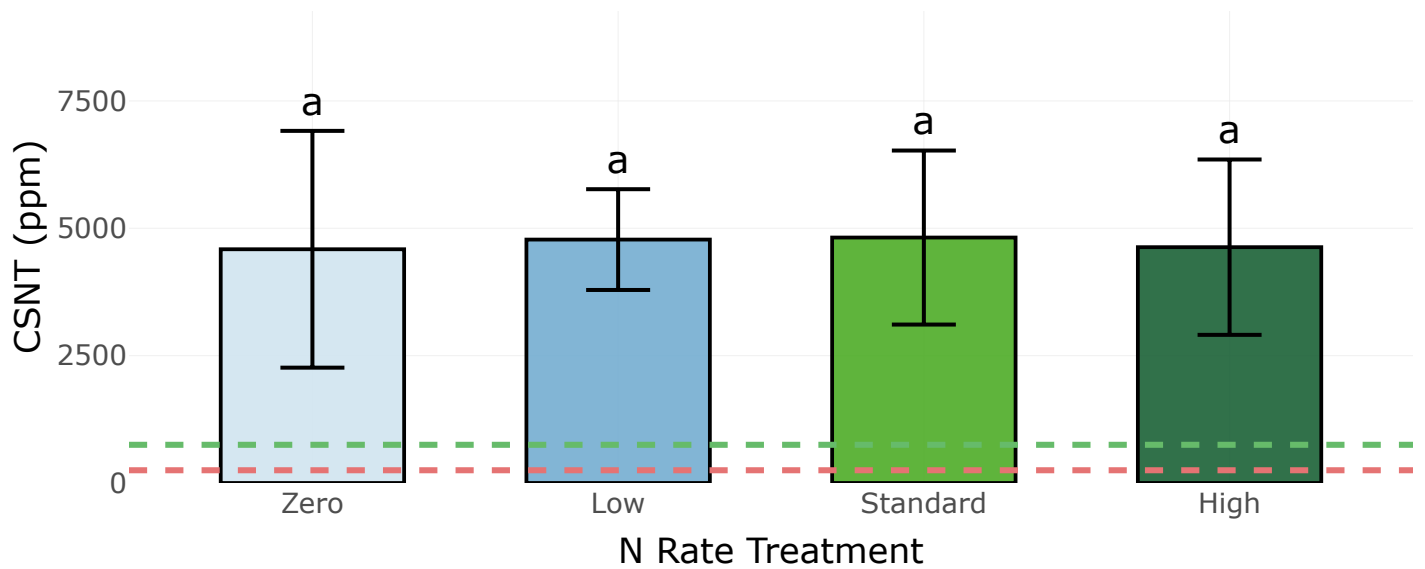
RCBD ANOVA — E and W samples averaged per plot

— No significant difference detected

Mean $\pm$ SE + Tukey

[Raw data by plot](#)

Cornstalk Nitrate (CSNT) – Mean $\pm$ SE by N Rate



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test** (sampled 04/30/2025, Ntrials field). The overall quality score was **57/100 (Medium)**. The biological indicators are the main area of concern: soil respiration, organic matter, active carbon, and soil protein index all came back Low. These are the indicators that reflect how active and alive your soil biology is, and right now, that biological engine is running at a lower capacity than ideal. In line with that, the strips that received no fertilizer nitrogen yielded less this season, which suggests the soil on its own did not supply all the nitrogen the crop needed. Cover cropping is a direct investment in improving exactly these biological scores over time, feeding the microbes, adding carbon, and building a soil that can supply nitrogen more consistently from its own internal processes.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: John Macauley
5815 Swan Hill Rd
Mt. Morris, NY
jcmacauley@gmail.com

Agricultural Service Provider:
Ristow
aristow@famland.org

Sample ID:
Field ID:
Date Sampled:
Given Soil Type:
Crops Grown:
Tillage:
Coordinates:

QBBB1116
Graveland N Trial
04/30/2025
Howard
SOY / WHT / COG
no till
Latitude: 42.72055556
Longitude: -77.81416667

Measured Soil Textural Class: **sandy loam**

Sand: **62%** - Silt: **27%** - Clay: **11%**

Group	Indicator	Value	Rating	Constraints
physical	Predicted Available Water Capacity	0.16	66	
physical	Surface Hardness	159	52	
physical	Subsurface Hardness	284	56	
physical	Aggregate Stability	23.3	33	
biological	Organic Matter	1.6	29	
	Soil Organic Carbon: 0.95 / Total Carbon: 0.97 / Total Nitrogen: 0.08			
biological	ACE Soil Protein Index	4.9	24	
biological	Soil Respiration	0.3	17	Soil Microbial Abundance and Activity
biological	Active Carbon	348	31	
chemical	Soil pH	6	83	
chemical	Extractable Phosphorus	7.3	100	
chemical	Extractable Potassium	177	100	
chemical	Additional Nutrients		88	
	Ca: 501.5 / Mg: 87.7 / S: 4.4			
	Al: 13.5 / B: 0.08 / Cu: 0.07			
	Fe: 1.6 / Mn: 3.9 / Zn: 0.6			

Overall Quality Score: **57** / Medium

Supplemental Soil Testing (see last page of report)

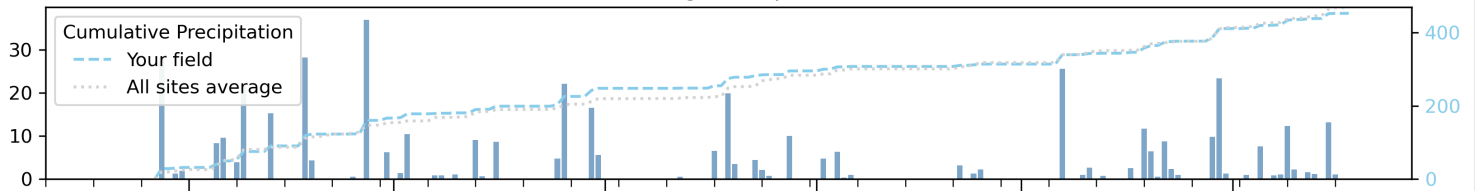
Soluble Salts (Conductivity)
0.34 mmhos/cm

[Open Full Soil Health Report \(PDF\)](#)

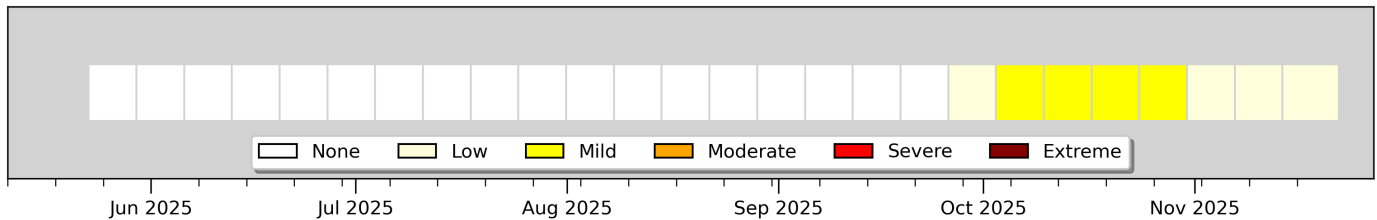
Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index from June through November 2025. Your field tracked closely with the all-site average through the summer months, with consistent rainfall from June through August and a cumulative total near the network average heading into September. A strong rain event in late June provided a good early push for the crop. The second half of the season brought a notable dry stretch, drought conditions escalated from Low in late September to Mild for most of October, lasting roughly four to five weeks before easing in November. This late-season dry period coincided with the end of grain fill and is worth keeping in mind when interpreting end-of-season stalk nitrate results, as dry conditions can concentrate nitrate in the stalk even when overall nitrogen supply is adequate.

Local vs All Site Average Precipitation 2025 (in mm)



Weekly Drought Index



Conclusions

CORN YIELD

Yield increased with nitrogen up to the standard rate: **146.2 bu/ac** with no nitrogen, 163.1 at 30 lb, **170** at 90 lb and 170.5 at 150 lb N/ac. The strips with no fertilizer yielded clearly less than the standard and high rates, while those two were the same. This season the crop needed fertilizer nitrogen to reach full yield, but not more than the standard rate.

NBI

NBI values were similar across all treatments (86–92), all within a healthy range. Adding more nitrogen did not improve leaf nitrogen status, indicating that in mid-July plants were well-supplied regardless of N rate.

CHLOROPHYLL

Chlorophyll was nearly identical across all N rates (46–48), confirming that leaf greenness was not affected by how much nitrogen was applied. At the time of the readings (mid-July), plants showed no signs of nitrogen stress even at zero applied N.

BIOMASS %N

Tissue nitrogen was consistent across treatments (4.1–4.2%), with no meaningful difference between the zero and high N rate strips. N rate had no detectable effect on how much nitrogen the plant accumulated at this growth stage.

CSNT

End-of-season stalk nitrate was very high on average in all treatments (4,588–4,818 ppm), well above the 750 ppm optimal threshold. Individual plots varied enormously, from about 140 to more than 10,000 ppm, so these averages are rough indications.

SOIL HEALTH

The overall soil health score was **57/100 (Medium)**, with low biological activity across the board. In line with this, the strips with no fertilizer nitrogen yielded less. Continued cover cropping could improve organic matter and microbial activity over time, even if a single-season nitrogen response is not yet visible.